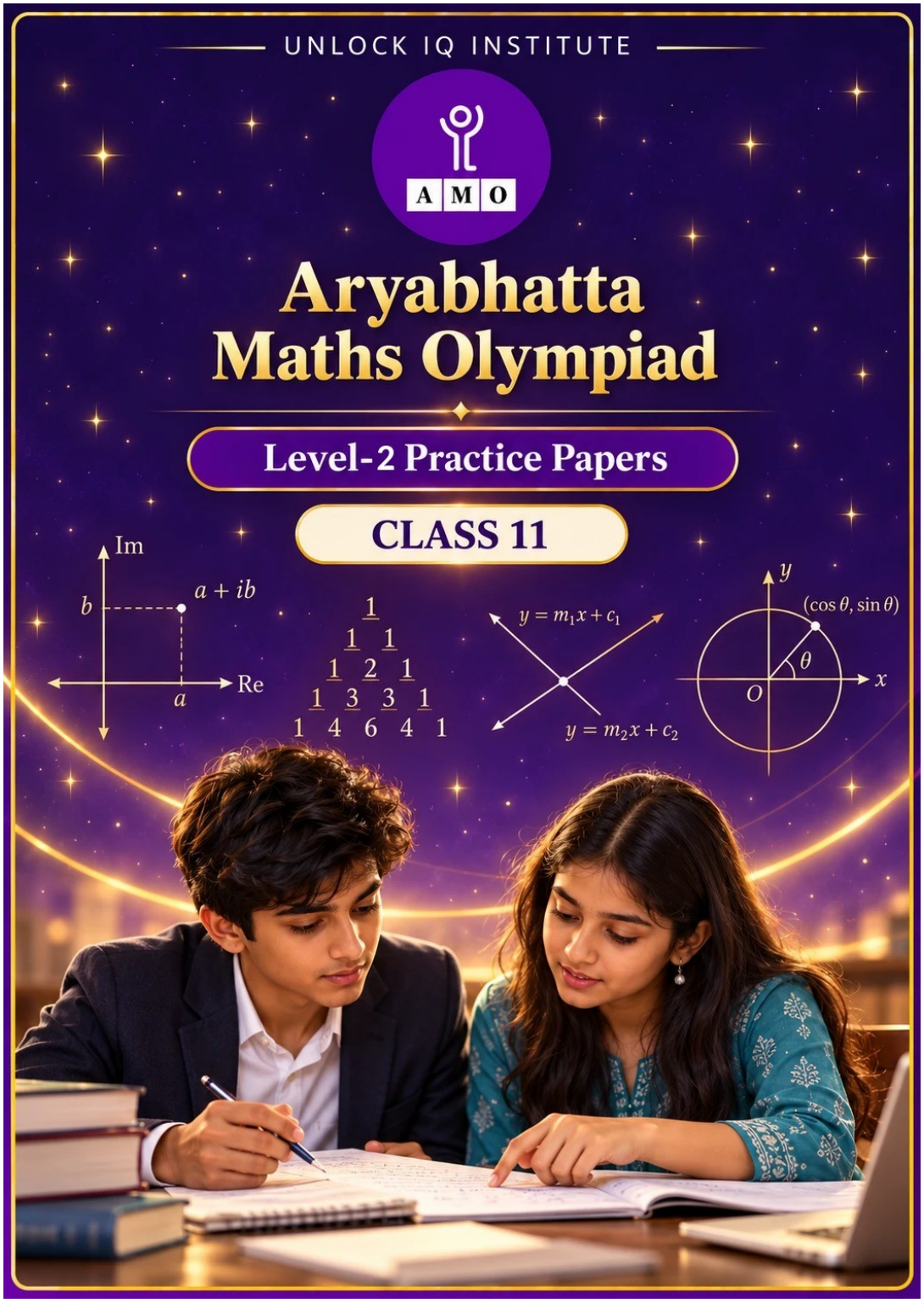


Set 2 · Practice Papers 6–10 · with answer keys & detailed solutions

Name: _____	**Class & Section:** _____
Roll No.: _____	**School:** _____
City: _____	**Date:** _____

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ARYABHATTA MATHS OLYMPIAD

Level 2

PRACTICE PAPER-6

CLASS 11

Total Questions	30 multiple-choice questions
Level	Level-2 (final round): every question is Hard
Time	45 minutes
Arrangement	Parts by chapter; the last Part has mixed problems joining two chapters
Negative Marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Algebra · Geometry · Trigonometry · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning

Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

For teachers

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2, 3 and 6 are not part of the Class 11 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

Part 1 — Number System (Syllabus Chapter 1)

Q1. Let $z = 3 + 4i$. What is the area of the triangle whose vertices are the points z, iz and $z + iz$ in the complex plane?

(A) 25 **(B)** 5 **(C)** 25/2 **(D)** 50

Q2. Let $w = (1 + i)/(1 - i)$. What is the value of $w + w^2 + w^3 + \ldots + w^{2026}$?

(A) $-1 + i$ **(B)** 0 **(C)** i **(D)** $-1 - i$

Q3. A complex number z satisfies $|z - 2| = |z - 4i|$ and $|z| = |z - 6|$. What is z^2?

(A) $18i$ **(B)** 18 **(C)** $-18i$ **(D)** $9 + 9i$

Part 2 — Algebra (Syllabus Chapter 4)

Q4. In an AP, the sum of the first 10 terms is 20 and the sum of the first 20 terms is 10. What is the sum of the first 30 terms?

(A) 0 **(B)** 30 **(C)** -30 **(D)** -10

Q5. What is the value of $1\cdot2 + 2\cdot2^2 + 3\cdot2^3 + \ldots + 10\cdot2^{10}$?

(A) 20482 **(B)** 18434 **(C)** 18432 **(D)** 9218

Q6. What is the term independent of x in the expansion of $(x + 2/x)^8$?

(A) 70 **(B)** 1120 **(C)** 560 **(D)** 2240

Q7. In the expansion of $(1 + x)^n$, the coefficients of x^4 and x^5 are equal. What is the coefficient of x^3?

(A) 84 **(B)** 120 **(C)** 56 **(D)** 36

Q8. What is the least value of $(a + b)(1/a + 4/b)$ for positive real numbers a and b?

(A) 8 **(B)** 5 **(C)** 4 **(D)** 9

Q9. How many points (x, y) with both coordinates integers satisfy $|x| + |y| \le 3$?

(A) 24 **(B)** 18 **(C)** 25 **(D)** 49

Part 3 — Geometry (Syllabus Chapter 5)

Q10. What is the orthocentre of the triangle with vertices (0, 0), (4, 0) and (1, 3)?

(A) (5/3, 1) **(B)** (2, 1) **(C)** (1, 1) **(D)** (1, 3/2)

Q11. Two lines through (3, 2) each make an angle of 45° with the line $x - 2y = 3$. What is the sum of their slopes?

(A) −8/3 **(B)** 3 **(C)** 1/2 **(D)** 8/3

Q12. The parallel lines $x + 2y = 4$ and $2x + 4y = k$ are $\sqrt{5}$ apart. What are the possible values of k?

(A) 13 or 3 **(B)** 9 or −1 **(C)** 18 only **(D)** 18 or −2

Part 4 — Trigonometry (Syllabus Chapter 7)

Q13. What is the value of $\cos 15° + \cos 75° + \cos 135°$?

(A) $(\sqrt{6} - \sqrt{2})/2$ **(B)** $\sqrt{6}/2$ **(C)** $(\sqrt{6} + \sqrt{2})/2$ **(D)** 0

Q14. If $\cos\theta = 1/3$, what is $\cos 4\theta$?

(A) −7/9 **(B)** −17/81 **(C)** 49/81 **(D)** 17/81

Q15. What is the value of $\sin(-420°)\cos 390° + \cos(-660°)\sin 330°$?

(A) 1 **(B)** 0 **(C)** −1 **(D)** −1/2

Part 5 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Q16. There are 10 points in a plane. Exactly 4 of them lie on one straight line, and 3 others lie on a second straight line. No other three of the points are in a line. How many triangles have their vertices among these points?

(A) 116 **(B)** 113 **(C)** 115 **(D)** 111

Q17. In how many ways can three different numbers be chosen from 1, 2, … , 20 so that their sum is even?

(A) 1140 **(B)** 570 **(C)** 120 **(D)** 450

Q18. A pair of dice is thrown 4 times. What is the probability of getting a doublet (both dice showing the same number) exactly twice?

(A) 1/36 **(B)** 25/1296 **(C)** 5/72 **(D)** 25/216

Q19. The variance of the first n natural numbers 1, 2, ... , n is 10. What is n?

(A) 10 **(B)** 11 **(C)** 12 **(D)** 9

Part 6 — Applied Mathematics (Syllabus Chapter 9)

Q20. Bhavna deposits ₹1,000 at the start of each year for 3 years in a scheme paying 10% per annum compound interest. What is the value at the end of the 3rd year?

(A) ₹3,310 **(B)** ₹3,641 **(C)** ₹3,300 **(D)** ₹3,993

Q21. The difference between compound interest and simple interest on a sum for 3 years at 10% per annum is ₹155. What is the sum?

(A) ₹15,500 **(B)** ₹1,550 **(C)** ₹5,000 **(D)** ₹6,200

Q22. Asha and Bhavin finish a job in 12 days, Bhavin and Charu in 15 days, and Charu and Asha in 20 days. How long would Charu take alone?

(A) 30 days **(B)** 20 days **(C)** 40 days **(D)** 60 days

Part 7 — Mathematical Reasoning (Syllabus Chapter 10)

Q23. What is the least n such that among any n integers there is always a non-empty group whose sum is divisible by 10?

(A) 10 **(B)** 11 **(C)** 9 **(D)** 19

Q24. Among a number of identical-looking coins, exactly one is heavier. With a two-pan balance, what is the largest number of coins for which 4 weighings are always enough to find the heavy coin?

(A) 64 **(B)** 81 **(C)** 27 **(D)** 16

Q25. What is the largest integer that divides $n(n + 1)(n + 2)(n + 3)$ for every positive integer n?

(A) 12 **(B)** 24 **(C)** 48 **(D)** 120

Part 8 — Mixed Problems (each joins two chapters)

Q26. A sequence has $a_1 = 1$, $a_2 = 3$ and $a_{n+1} = 3a_n - 2a_{n-1}$ for $n \geq 2$. What is a_{10}?

(A) 1023 **(B)** 511 **(C)** 1024 **(D)** 2047

Q27. A fair coin is tossed 10 times. What is the probability of getting more heads than tails?

(A) 1/2 **(B)** 63/256 **(C)** 193/512 **(D)** 319/512

Q28. What is the sum of the first 10 terms of $(1 + i) + (2 + 3i) + (3 + 5i) + \ldots$?

(A) 55 + 95i **(B)** 100 + 55i **(C)** 45 + 100i **(D)** 55 + 100i

Q29. Two lines pass through the point (1, 1) and each makes an angle of 15° with the line $y = x$. What is the product of their slopes?

(A) 1 **(B)** $\sqrt{3}$ **(C)** 2 **(D)** 4

Q30. Three vertices of a regular hexagon are chosen at random. What is the probability that they form an equilateral triangle?

(A) 1/20 **(B)** 1/10 **(C)** 1/5 **(D)** 1/3

Answer Sheet AMO Level 2 · Class 11 · Practice Paper 6

Name: **School:** **Roll No.:**

Shade or tick one circle for each question.

1.	(A)	(B)	(C)	(D)	11.	(A)	(B)	(C)	(D)	21.	(A)	(B)	(C)	(D)
2.	(A)	(B)	(C)	(D)	12.	(A)	(B)	(C)	(D)	22.	(A)	(B)	(C)	(D)
3.	(A)	(B)	(C)	(D)	13.	(A)	(B)	(C)	(D)	23.	(A)	(B)	(C)	(D)
4.	(A)	(B)	(C)	(D)	14.	(A)	(B)	(C)	(D)	24.	(A)	(B)	(C)	(D)
5.	(A)	(B)	(C)	(D)	15.	(A)	(B)	(C)	(D)	25.	(A)	(B)	(C)	(D)
6.	(A)	(B)	(C)	(D)	16.	(A)	(B)	(C)	(D)	26.	(A)	(B)	(C)	(D)
7.	(A)	(B)	(C)	(D)	17.	(A)	(B)	(C)	(D)	27.	(A)	(B)	(C)	(D)
8.	(A)	(B)	(C)	(D)	18.	(A)	(B)	(C)	(D)	28.	(A)	(B)	(C)	(D)
9.	(A)	(B)	(C)	(D)	19.	(A)	(B)	(C)	(D)	29.	(A)	(B)	(C)	(D)
10.	(A)	(B)	(C)	(D)	20.	(A)	(B)	(C)	(D)	30.	(A)	(B)	(C)	(D)

For the parent or teacher

Correct: _____ / 30

Answer Key and Solutions AMO Level 2 · Class 11 · Practice Paper 6

Quick answers

1 **C**	6 **B**	11 **D**	16 **C**	21 **C**	26 **A**
2 **A**	7 **A**	12 **D**	17 **B**	22 **D**	27 **C**
3 **A**	8 **D**	13 **A**	18 **D**	23 **A**	28 **D**
4 **C**	9 **C**	14 **D**	19 **B**	24 **B**	29 **A**
5 **B**	10 **C**	15 **C**	20 **B**	25 **B**	30 **B**

Solutions

Q	Part · level	Answer	Why?
1	Part 1 · Ch1 · Hard	**(C)** 25/2	***Hint:*** *Multiplying by i turns a point through 90° about the origin, keeping its distance.* iz = −4 + 3i and z + iz = −1 + 7i. The sides from z + iz to z and to iz are iz and z, which are perpendicular and each of length 5. So the triangle is right-angled with legs 5 and 5: area 25/2. (Shoelace on (3, 4), (−4, 3), (−1, 7) gives the same.)
2	Part 1 · Ch1 · Hard	**(A)** −1 + i	***Hint:*** *First simplify w. Its powers repeat every 4 terms.* $w = (1 + i)^2/2 = i$. Every block $i + i^2 + i^3 + i^4$ adds to 0. 2026 = 4 × 506 + 2, so the sum is $i + i^2 = -1 + i$.
3	Part 1 · Ch1 · Hard	**(A)** 18i	***Hint:*** *Each condition puts z on a perpendicular bisector. Write z = x + iy.* $\|z\| = \|z - 6\|$ gives x = 3. $\|z - 2\| = \|z - 4i\|$ gives $(x - 2)^2 + y^2 = x^2 + (y - 4)^2$, so 8y = 4x + 12 and y = 3. So z = 3 + 3i and $z^2 = 9 - 9 + 18i = 18i$.
4	Part 2 · Ch4 · Hard	**(C)** −30	***Hint:*** *Write $S_n = An^2 + Bn$ and use the two facts to find A and B.* 100A + 10B = 20 and 400A + 20B = 10 give A = −0.15 and B = 3.5. S_{30} = 900(−0.15) + 30(3.5) = −135 + 105 = −30. (In general, if $S_p = q$ and S_q = p, then $S_{p+}q = -(p + q)$.)
5	Part 2 · Ch4 · Hard	**(B)** 18434	***Hint:*** *Let S be the sum. Work out 2S − S; the middle terms form a GP.* $2S - S = 10 \cdot 2^{11} - (2 + 2^2 + \ldots + 2^{10}) = 20480 - 2046 = 18434$. (In general $\Sigma k \cdot 2^k = (n - 1)2^{n+1} + 2$.)
6	Part 2 · Ch4 · Hard	**(B)** 1120	***Hint:*** *The general term is $C(8, r)\, x^{8-r}\, (2/x)^r$. Find r for which the power of x is 0.* The power of x is 8 − 2r = 0, so r = 4. The term is $C(8, 4) \times 2^4 = 70 \times 16 = 1120$. (70 forgets the factor 2^4.)
7	Part 2 · Ch4 · Hard	**(A)** 84	***Hint:*** *C(n, 4) = C(n, 5) happens when 4 + 5 = n.* C(n, 4) = C(n, 5) means n = 9. The coefficient of x^3 is C(9, 3) = 84.
8	Part 2 · Ch4 · Hard	**(D)** 9	***Hint:*** *Expand: 1 + 4 + b/a + 4a/b. Apply AM ≥ GM to the last two terms.* (a + b)(1/a + 4/b) = 5 + b/a + 4a/b ≥ 5 + 2√4 = 9, with equality when b = 2a.
9	Part 2 · Ch4 · Hard	**(C)** 25	***Hint:*** *Count row by row: for y = 0, ±1, ±2, ±3.* y = 0: x from −3 to 3 (7). y = ±1: 5 each. y = ±2: 3 each. y = ±3: 1 each. Total 7 + 10 + 6 + 2 = 25.

Q	Part · level	Answer	Why?
10	Part 3 · Ch5 · Hard	**(C)** (1, 1)	***Hint:*** *The altitude from (1, 3) is vertical. Find one more altitude.* The altitude from (1, 3) to the base on the x-axis is x = 1. The side from (0, 0) to (1, 3) has slope 3, so the altitude from (4, 0) has slope −1/3: y = −(x − 4)/3. At x = 1, y = 1. Orthocentre (1, 1). ((5/3, 1) is the centroid.)
11	Part 3 · Ch5 · Hard	**(D)** 8/3	***Hint:*** *Solve $\lvert(m - ½)/(1 + m/2)\rvert = 1$.* (m − ½)/(1 + m/2) = 1 gives m = 3. (m − ½)/(1 + m/2) = −1 gives m = −1/3. Sum = 3 − 1/3 = 8/3. The lines are 3x − y = 7 and x + 3y = 9.
12	Part 3 · Ch5 · Hard	**(D)** 18 or −2	***Hint:*** *Write the second line as x + 2y = k/2.* Distance = $\lvert4 - k/2\rvert/\sqrt{5} = \sqrt{5}$, so $\lvert4 - k/2\rvert = 5$. Then k/2 = 9 or −1, giving k = 18 or −2.
13	Part 4 · Ch7 · Hard	**(A)** (√6 − √2)/2	***Hint:*** *Use cos A + cos B = 2 cos((A + B)/2) cos((A − B)/2) on the first two terms.* cos 15° + cos 75° = 2 cos 45° cos 30° = 2 × (1/√2) × (√3/2) = √6/2. cos 135° = −1/√2 = −√2/2. Sum = (√6 − √2)/2.
14	Part 4 · Ch7 · Hard	**(D)** 17/81	***Hint:*** *Use $\cos 2x = 2\cos^2 x - 1$ twice.* cos 2θ = 2/9 − 1 = −7/9. cos 4θ = 2(49/81) − 1 = 17/81.
15	Part 4 · Ch7 · Hard	**(C)** −1	***Hint:*** *Reduce each angle: −420° → −60°, 390° → 30°, −660° → 60°, 330° → −30°.* sin(−420°) = −sin 60° = −√3/2. cos 390° = cos 30° = √3/2. cos(−660°) = cos 60° = 1/2. sin 330° = −1/2. Sum = −3/4 − 1/4 = −1.
16	Part 5 · Ch8 · Hard	**(C)** 115	***Hint:*** *Count all choices of 3 points, then remove the choices that lie on a line.* All choices: C(10, 3) = 120. Collinear choices: C(4, 3) = 4 on the first line and C(3, 3) = 1 on the second. Triangles: 120 − 4 − 1 = 115. (116 forgets the second line.)
17	Part 5 · Ch8 · Hard	**(B)** 570	***Hint:*** *The sum is even with three evens, or with one even and two odds.* Three evens: C(10, 3) = 120. One even and two odds: 10 × C(10, 2) = 450. Total 570 (exactly half of C(20, 3) = 1140).
18	Part 5 · Ch8 · Hard	**(D)** 25/216	***Hint:*** *A doublet has probability 1/6 on each throw of the pair.* $P = C(4, 2)(1/6)^2(5/6)^2 = 6 \times 25/1296 = 150/1296 = 25/216$.
19	Part 5 · Ch8 · Hard	**(B)** 11	***Hint:*** *The variance of 1, 2, … , n is $(n^2 - 1)/12$.* $(n^2 - 1)/12 = 10$ gives $n^2 = 121$, so n = 11.
20	Part 6 · Ch9 · Hard	**(B)** ₹3,641	***Hint:*** *The deposits grow for 3, 2 and 1 years respectively.* $1{,}000(1.1^3 + 1.1^2 + 1.1) = 1{,}000(1.331 + 1.21 + 1.1) =$ ₹3,641. (₹3,310 assumes deposits at the end of each year.)
21	Part 6 · Ch9 · Hard	**(C)** ₹5,000	***Hint:*** *For 3 years at rate r, CI − SI = $P(3r^2 + r^3)$.* $3r^2 + r^3 = 3(0.01) + 0.001 = 0.031$. P = 155 ÷ 0.031 = ₹5,000.
22	Part 6 · Ch9 · Hard	**(D)** 60 days	***Hint:*** *Add the three pair rates: that is twice the rate of all three together.* 1/12 + 1/15 + 1/20 = 12/60 = 1/5 is twice the combined rate, so together they do 1/10 a day. Charu = 1/10 − 1/12 (Asha and Bhavin) = 1/60. Charu takes 60 days.
23	Part 7 · Ch10 · Hard	**(A)** 10	***Hint:*** *Look at the partial sums a_1, $a_1 + a_2$, … , $a_1 + … + a_n$ modulo 10.* With 10 integers there are 10 partial sums. If one is 0 mod 10, done; otherwise two of them share a remainder (only 9 non-zero remainders), and the terms between them sum to a multiple of 10. Nine 1s show that 9 is not enough.

Q	Part · level	Answer	Why?
24	Part 7 · Ch10 · Hard	**(B)** 81	***Hint:*** *Each weighing has 3 outcomes: left heavier, right heavier or balance.* Each weighing splits the possibilities into at most 3 groups, so 4 weighings can tell apart at most $3^4 = 81$ coins. With 81 coins, weigh 27 against 27 and keep the heavy group (or the 27 left out); repeat with 9, 3, 1. So 81. ($16 = 2^4$ treats each weighing as having only two outcomes.)
25	Part 7 · Ch10 · Hard	**(B)** 24	***Hint:*** *The product is 4! × C(n + 3, 4). Check n = 1.* $n(n + 1)(n + 2)(n + 3) = 24 \times C(n + 3, 4)$, so 24 always divides it. For n = 1 the product is exactly 24, so no larger number works.
26	Part 8 · Mixed · Ch4 + Ch10 · Hard	**(A)** 1023	***Hint:*** *Look at the differences $a_{n+1} - a_n$.* $a_{n+1} - a_n = 2(a_n - a_{n-1})$, so the differences are 2, 4, 8, … The formula $a_n = 2^n - 1$ fits a_1 and a_2, and induction carries it forward: $3(2^n - 1) - 2(2^{n-1} - 1) = 2^{n+1} - 1$. So $a_{10} = 1023$.
27	Part 8 · Mixed · Ch4 + Ch8 · Hard	**(C)** 193/512	***Hint:*** *By symmetry, P(more heads) = P(more tails). Remove the case of equal numbers.* P(5 heads) = C(10, 5)/1024 = 252/1024. The rest splits equally: (1 − 252/1024)/2 = 772/2048 = 193/512.
28	Part 8 · Mixed · Ch1 + Ch4 · Hard	**(D)** 55 + 100i	***Hint:*** *The real parts and the imaginary parts form separate APs.* Real parts 1, 2, … , 10 add up to 55. Imaginary parts 1, 3, … , 19 add up to $10^2 = 100$. Sum = 55 + 100i.
29	Part 8 · Mixed · Ch5 + Ch7 · Hard	**(A)** 1	***Hint:*** *y = x has inclination 45°. The two lines have inclinations 45° ± 15°.* The inclinations are 60° and 30°, so the slopes are tan 60° = √3 and tan 30° = 1/√3. Their product is 1.
30	Part 8 · Mixed · Ch8 + Ch10 · Hard	**(B)** 1/10	***Hint:*** *Only alternate vertices give an equilateral triangle.* There are C(6, 3) = 20 choices. Only vertices 1, 3, 5 and 2, 4, 6 give equilateral triangles. P = 2/20 = 1/10.

ARYABHATTA MATHS OLYMPIAD

Level 2

PRACTICE PAPER-7

CLASS 11

Total Questions	30 multiple-choice questions
Level	Level-2 (final round): every question is Hard
Time	45 minutes
Arrangement	Parts by chapter; the last Part has mixed problems joining two chapters
Negative Marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Algebra · Geometry · Trigonometry · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning

Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
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For teachers

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2, 3 and 6 are not part of the Class 11 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

Part 1 — Number System (Syllabus Chapter 1)

Q1. Real numbers x and y satisfy $(x + iy)(2 - 3i) = 4 + i$. What is $x + y$?

(A) 19/13 **(B)** 1 **(C)** −9/13 **(D)** 5/13

Q2. A complex number z with positive imaginary part satisfies $z \cdot \bar{z} = 25$ and $z + \bar{z} = 6$. What is z^4?

(A) −527 − 336i **(B)** 527 + 336i **(C)** −527 + 336i **(D)** 625

Q3. How many complex numbers $z = a + ib$, with a and b integers, satisfy $|z| = 5$?

(A) 4 **(B)** 8 **(C)** 6 **(D)** 12

Part 2 — Algebra (Syllabus Chapter 4)

Q4. What is the greatest value of the sum of the first n terms of the AP 100, 97, 94, … ?

(A) 1683 **(B)** 1700 **(C)** 1717 **(D)** 1750

Q5. An infinite GP has sum 20, and the sum of the squares of its terms is 100. What is its first term?

(A) 5 **(B)** 8 **(C)** 10 **(D)** 12

Q6. What is the sum of all natural numbers less than 200 that are divisible by neither 3 nor 5?

(A) 9367 **(B)** 11097 **(C)** 10732 **(D)** 10632

Q7. What is the sum to infinity of $1 + 2/3 + 3/9 + 4/27 + \ldots$?

(A) 3/2 **(B)** 3 **(C)** 4/3 **(D)** 9/4

Q8. Positive real numbers a, b and c satisfy $a + b + c = 12$. What is the greatest possible value of ab^2c?

(A) 256 **(B)** 64 **(C)** 324 **(D)** 432

Q9. Using the binomial theorem, what is $(0.99)^5$ correct to three decimal places?

(A) 0.950 **(B)** 0.995 **(C)** 0.941 **(D)** 0.951

Part 3 — Geometry (Syllabus Chapter 5)

Q10. What is the area of the triangle bounded by the lines $y = x$, $y = -x$ and $3x + 4y = 14$?

(A) 56 **(B)** 14 **(C)** 28 **(D)** 32

Q11. What is the incentre of the triangle with vertices (0, 0), (6, 0) and (0, 8)?

(A) (1, 1) **(B)** (2, 8/3) **(C)** (3, 4) **(D)** (2, 2)

Q12. The line $3x + 4y = 12$ is turned through 90° about the point where it meets the x-axis. Where does the new line meet the y-axis?

(A) $(0, -16/3)$ **(B)** $(0, 16/3)$ **(C)** $(0, -4)$ **(D)** (0, 3)

Part 4 — Trigonometry (Syllabus Chapter 7)

Q13. How many different integer values can $5 \sin x + 12 \cos x$ take as x varies over all real numbers?

(A) 13 **(B)** 27 **(C)** 25 **(D)** 17

Q14. If $\sin A + \sin B = 1/2$ and $\cos A + \cos B = \sqrt{3}/2$, what is $\cos(A + B)$?

(A) 1/2 **(B)** $\sqrt{3}/2$ **(C)** $-1/2$ **(D)** 0

Q15. θ is in the second quadrant ($90° < \theta < 180°$) and $\tan \theta = -4/3$. What is $\sin(\theta/2)$?

(A) $2/\sqrt{5}$ **(B)** $1/\sqrt{5}$ **(C)** $-2/\sqrt{5}$ **(D)** 4/5

Part 5 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Q16. Six-digit numbers are formed using each of the digits 1, 2 and 3 exactly twice. In how many of them are no two equal digits next to each other?

(A) 90 **(B)** 30 **(C)** 48 **(D)** 36

Q17. A path goes from (0, 0) to (4, 4) in unit steps to the right or up. How many such paths pass through neither (1, 1) nor (3, 3)?

(A) 34 **(B)** 20 **(C)** 6 **(D)** 14

Q18. A fair coin is tossed 10 times. What is the probability of getting at least 8 heads?

(A) 45/1024 **(B)** 1/128 **(C)** 11/1024 **(D)** 7/128

Q19. What is the standard deviation of 3, 6, 9, 12 and 15?

(A) $3\sqrt{2}$ **(B)** 18 **(C)** $\sqrt{6}$ **(D)** 9

Part 6 — Applied Mathematics (Syllabus Chapter 9)

Q20. A town of 20,000 people grows by 5% in the first year and by 4% in the second, then falls by 10% in the third. What is the population after three years?

(A) 19,800 **(B)** 21,840 **(C)** 19,600 **(D)** 19,656

Q21. A workshop has fixed monthly costs of ₹1,20,000. Each lamp costs ₹300 to make and is sold for ₹450 plus 18% GST, which the workshop collects and pays to the government. How many lamps must it sell in a month to make a profit of ₹30,000?

(A) 800 **(B)** 1,000 **(C)** 650 **(D)** 1,250

Q22. A boat moves at 9 km/h in still water. It takes twice as long to go a distance upstream as to go the same distance downstream. How long does a trip of 24 km downstream and 24 km back take?

(A) 5⅓ h **(B)** 6 h **(C)** 4 h **(D)** 8 h

Part 7 — Mathematical Reasoning (Syllabus Chapter 10)

Q23. At a party of 9 people, can every person shake hands with exactly 3 others?

(A) Yes, by arranging them in three groups of three

(B) Yes, by seating them in a circle

(C) No, because the total of all the handshake counts would be odd

(D) No, because 9 is not a multiple of 4

Q24. What is the largest number of kings that can be placed on an 8 × 8 chessboard so that no two attack each other?

(A) 32 **(B)** 16 **(C)** 64 **(D)** 9

Q25. Some positive integers add up to 10. What is the largest possible value of their product?

(A) 32 **(B)** 25 **(C)** 30 **(D)** 36

Part 8 — Mixed Problems (each joins two chapters)

Q26. What is the least n for which $1^3 + 2^3 + \ldots + n^3 > 10{,}000$?

(A) 13 **(B)** 10 **(C)** 14 **(D)** 22

Q27. What is the coefficient of x^3 in the expansion of $(1 + x + x^2)^5$?

(A) 10 **(B)** 35 **(C)** 30 **(D)** 20

Q28. What is the sum to infinity of $1 + i/2 + (i/2)^2 + (i/2)^3 + \ldots$?

(A) $(4 + 2i)/5$ **(B)** 2 **(C)** $(4 - 2i)/5$ **(D)** $(2 + 4i)/5$

Q29. A point moves 4 units from (1, 2) along a line making 60° with the positive x-axis, going up and to the right. Where does it end?

(A) $(1 + 2\sqrt{3}, 4)$ **(B)** $(3, 2 + 2\sqrt{3})$ **(C)** $(5, 2 + 4\sqrt{3})$ **(D)** $(3, 2 + 4\sqrt{3})$

Q30. Numbers are chosen from 1, 2, … , 8. What is the least number that must be chosen to be sure that two of them add up to 9?

(A) 5 **(B)** 4 **(C)** 6 **(D)** 8

Answer Sheet AMO Level 2 · Class 11 · Practice Paper 7

Name: **School:** **Roll No.:**

Shade or tick one circle for each question.

1.	A	B	C	D	11.	A	B	C	D	21.	A	B	C	D
2.	A	B	C	D	12.	A	B	C	D	22.	A	B	C	D
3.	A	B	C	D	13.	A	B	C	D	23.	A	B	C	D
4.	A	B	C	D	14.	A	B	C	D	24.	A	B	C	D
5.	A	B	C	D	15.	A	B	C	D	25.	A	B	C	D
6.	A	B	C	D	16.	A	B	C	D	26.	A	B	C	D
7.	A	B	C	D	17.	A	B	C	D	27.	A	B	C	D
8.	A	B	C	D	18.	A	B	C	D	28.	A	B	C	D
9.	A	B	C	D	19.	A	B	C	D	29.	A	B	C	D
10.	A	B	C	D	20.	A	B	C	D	30.	A	B	C	D

For the parent or teacher

Correct: _____ / 30

Answer Key and Solutions AMO Level 2 · Class 11 · Practice Paper 7

Quick answers

1 **A**	6 **C**	11 **D**	16 **B**	21 **B**	26 **C**
2 **A**	7 **D**	12 **A**	17 **D**	22 **B**	27 **C**
3 **D**	8 **C**	13 **B**	18 **D**	23 **C**	28 **A**
4 **C**	9 **D**	14 **A**	19 **A**	24 **B**	29 **B**
5 **B**	10 **C**	15 **A**	20 **D**	25 **D**	30 **A**

Solutions

Q	Part · level	Answer	Why?
1	Part 1 · Ch1 · Hard	**(A)** 19/13	***Hint:*** *$x + iy = (4 + i)/(2 − 3i)$. Multiply top and bottom by $2 + 3i$.* $x + iy = (4 + i)(2 + 3i)/13 = (8 + 12i + 2i − 3)/13 = (5 + 14i)/13$. So $x = 5/13$, $y = 14/13$ and $x + y = 19/13$.
2	Part 1 · Ch1 · Hard	**(A)** −527 − 336i	***Hint:*** *Find z first, then square twice.* Re $z = 3$ and $\|z\| = 5$, so $z = 3 + 4i$. $z^2 = −7 + 24i$. $z^4 = (−7 + 24i)^2 = 49 − 576 − 336i = −527 − 336i$. ($625 = \|z\|^4$.)
3	Part 1 · Ch1 · Hard	**(D)** 12	***Hint:*** *Find all integer solutions of $a^2 + b^2 = 25$, with signs.* $a^2 + b^2 = 25$: (±5, 0), (0, ±5) give 4; (±3, ±4) give 4; (±4, ±3) give 4. Total 12. (4 counts only the points on the axes.)
4	Part 2 · Ch4 · Hard	**(C)** 1717	***Hint:*** *Keep adding terms only while they are positive.* $T_n = 103 − 3n > 0$ for $n \le 34$ ($T_{34} = 1$, $T_{35} = −2$). The greatest sum is $S_{34} = 34(100 + 1)/2 = 1717$.
5	Part 2 · Ch4 · Hard	**(B)** 8	***Hint:*** *$a/(1 − r) = 20$ and $a^2/(1 − r^2) = 100$. Divide the second by the first.* Dividing: $a/(1 + r) = 5$. With $a = 20(1 − r)$: $20(1 − r) = 5(1 + r)$, so $r = 3/5$ and $a = 20 × 2/5 = 8$.
6	Part 2 · Ch4 · Hard	**(C)** 10732	***Hint:*** *Start from $1 + 2 + … + 199$ and use inclusion–exclusion with three APs.* 1 to 199: 19900. Multiples of 3 (3 to 198): 6633. Multiples of 5 (5 to 195): 3900. Multiples of 15 (15 to 195): 1365. Answer: 19900 − 6633 − 3900 + 1365 = 10732.
7	Part 2 · Ch4 · Hard	**(D)** 9/4	***Hint:*** *Let S be the sum and work out $S − S/3$.* $S − S/3 = 1 + 1/3 + 1/9 + … = 3/2$. So $(2/3)S = 3/2$ and $S = 9/4$.
8	Part 2 · Ch4 · Hard	**(C)** 324	***Hint:*** *Apply AM ≥ GM to the four numbers a, $b/2$, $b/2$ and c.* $a + b/2 + b/2 + c = 12$, so $a(b/2)(b/2)c \le (12/4)^4 = 81$. Hence $ab^2c \le 4 × 81 = 324$, reached at $a = c = 3$, $b = 6$.
9	Part 2 · Ch4 · Hard	**(D)** 0.951	***Hint:*** *$(1 − 0.01)^5 = 1 − 5(0.01) + 10(0.01)^2 − …$* $(1 − 0.01)^5 = 1 − 0.05 + 0.001 − 0.00001 + … = 0.95099…$, which is 0.951 to three decimal places.
10	Part 3 · Ch5 · Hard	**(C)** 28	***Hint:*** *Find the two corners on $3x + 4y = 14$; the third corner is the origin.* $y = x$ meets $3x + 4y = 14$ at (2, 2). $y = −x$ meets it where $3x − 4x = 14$, at (−14, 14). Area $= ½\|2 × 14 − (−14) × 2\| = ½ × 56 = 28$.

Q	Part · level	Answer	Why?
11	Part 3 · Ch5 · Hard	**(D)** (2, 2)	***Hint:*** *For a right triangle with legs a, b and hypotenuse c, the inradius is (a + b − c)/2.* Legs 6 and 8, hypotenuse 10, so r = (6 + 8 − 10)/2 = 2. The incircle touches both axes, so its centre is (2, 2). ((2, 8/3) is the centroid and (3, 4) the circumcentre.)
12	Part 3 · Ch5 · Hard	**(A)** (0, −16/3)	***Hint:*** *The new line passes through (4, 0) and is perpendicular to the old one.* The old line meets the x-axis at (4, 0). The new line is perpendicular: 4x − 3y = c through (4, 0), so 4x − 3y = 16. At x = 0, y = −16/3.
13	Part 4 · Ch7 · Hard	**(B)** 27	***Hint:*** *a sin x + b cos x takes every value from $-\sqrt{(a^2+b^2)}$ to $\sqrt{(a^2+b^2)}$.* 5 sin x + 12 cos x = 13 sin(x + α) with tan α = 12/5, so it takes every value from −13 to 13. The integers −13 to 13 number 27.
14	Part 4 · Ch7 · Hard	**(A)** 1/2	***Hint:*** *Change both sums into products. Dividing gives tan((A + B)/2).* 2 sin((A + B)/2) cos((A − B)/2) = 1/2 and 2 cos((A + B)/2) cos((A − B)/2) = √3/2. Dividing: tan((A + B)/2) = 1/√3 = t. cos(A + B) = $(1 - t^2)/(1 + t^2)$ = (2/3)/(4/3) = 1/2.
15	Part 4 · Ch7 · Hard	**(A)** 2/√5	***Hint:*** *cos θ = −3/5. Use $\sin^2(\theta/2) = (1 - \cos\theta)/2$ and decide the sign.* cos θ = −3/5, so $\sin^2(\theta/2)$ = (1 + 3/5)/2 = 4/5. Since 45° < θ/2 < 90°, sin(θ/2) > 0: sin(θ/2) = 2/√5.
16	Part 5 · Ch8 · Hard	**(B)** 30	***Hint:*** *Use inclusion–exclusion on the pairs 11, 22 and 33 being together.* All: 6!/(2!2!2!) = 90. One given pair together: 5!/(2!2!) = 30, three such: 90. Two pairs together: 4!/2! = 12, three such: 36. All three: 3! = 6. No pair together: 90 − 90 + 36 − 6 = 30.
17	Part 5 · Ch8 · Hard	**(D)** 14	***Hint:*** *Use inclusion–exclusion on the paths through (1, 1) and through (3, 3).* All paths: C(8, 4) = 70. Through (1, 1): 2 × C(6, 3) = 40. Through (3, 3): C(6, 3) × 2 = 40. Through both: 2 × C(4, 2) × 2 = 24. Avoiding both: 70 − 40 − 40 + 24 = 14.
18	Part 5 · Ch8 · Hard	**(D)** 7/128	***Hint:*** *Add P(8), P(9) and P(10).* $(C(10, 8) + C(10, 9) + C(10, 10))/2^{10}$ = (45 + 10 + 1)/1024 = 56/1024 = 7/128.
19	Part 5 · Ch8 · Hard	**(A)** 3√2	***Hint:*** *These are 3 × (1, 2, 3, 4, 5). The variance of 1 to 5 is 2.* Variance of 1 to 5 = (25 − 1)/12 = 2. Multiplying by 3 multiplies the variance by 9: 18. SD = √18 = 3√2.
20	Part 6 · Ch9 · Hard	**(D)** 19,656	***Hint:*** *Multiply by 1.05, 1.04 and 0.9.* 20,000 × 1.05 × 1.04 × 0.9 = 19,656. (Adding the rates, −1%, gives 19,800.)
21	Part 6 · Ch9 · Hard	**(B)** 1,000	***Hint:*** *GST collected is passed on, so it is not income. Find the profit on each lamp.* Each lamp earns 450 − 300 = ₹150 towards costs and profit; the GST is paid over. Needed: 1,20,000 + 30,000 = ₹1,50,000, so 1,50,000 ÷ 150 = 1,000 lamps. (800 only breaks even; 650 wrongly counts the GST as income.)
22	Part 6 · Ch9 · Hard	**(B)** 6 h	***Hint:*** *First find the stream speed from 9 + s = 2(9 − s).* 9 + s = 2(9 − s) gives s = 3. Downstream 12 km/h: 2 h. Upstream 6 km/h: 4 h. Total 6 h. (5⅓ h uses 9 km/h both ways.)

Q	Part · level	Answer	Why?
23	Part 7 · Ch10 · Hard	**(C)** No, because the total of all the handshake counts would be odd	***Hint:*** *Each handshake is counted twice in the total of everyone's counts.* The sum of everyone's handshake counts is twice the number of handshakes, so it must be even. But 9 × 3 = 27 is odd. So it is impossible. (Three groups of three would give each person only 2 handshakes.)
24	Part 7 · Ch10 · Hard	**(B)** 16	***Hint:*** *Split the board into 2 × 2 blocks.* Each 2 × 2 block can hold at most one king (any two squares in it touch). There are 16 blocks, so at most 16. Kings on the squares with both coordinates odd achieve 16.
25	Part 7 · Ch10 · Hard	**(D)** 36	***Hint:*** *Replacing any part 4 or more by 2 and (part − 2) does not lower the product. Prefer 3s.* Use as many 3s as possible: 10 = 3 + 3 + 4 (or 3 + 3 + 2 + 2), product 36. (2^5 = 32 and 5 × 5 = 25 are smaller.)
26	Part 8 · Mixed · Ch4 + Ch10 · Hard	**(C)** 14	***Hint:*** *Prove by induction that $1^3 + \ldots + n^3 = [n(n + 1)/2]^2$.* We need n(n + 1)/2 > 100, that is n(n + 1) > 200. 13 × 14 = 182 is too small; 14 × 15 = 210 works. So n = 14.
27	Part 8 · Mixed · Ch4 + Ch8 · Hard	**(C)** 30	***Hint:*** *Choose how many of the 5 brackets give x and how many give x^2.* x^3 comes from three brackets giving x (C(5, 3) = 10 ways) or from one x^2 and one x (5 × 4 = 20 ways). Total 10 + 20 = 30.
28	Part 8 · Mixed · Ch1 + Ch4 · Hard	**(A)** (4 + 2i)/5	***Hint:*** *The GP formula a/(1 − r) still works, since $\lvert r \rvert = 1/2 < 1$.* Sum = 1/(1 − i/2) = 2/(2 − i) = 2(2 + i)/5 = (4 + 2i)/5.
29	Part 8 · Mixed · Ch5 + Ch7 · Hard	**(B)** (3, 2 + 2√3)	***Hint:*** *The change in x is 4 cos 60° and the change in y is 4 sin 60°.* Δx = 4 × ½ = 2 and Δy = 4 × √3/2 = 2√3. The end point is (3, 2 + 2√3).
30	Part 8 · Mixed · Ch8 + Ch10 · Hard	**(A)** 5	***Hint:*** *Group the numbers into pairs that add up to 9.* Pairs: (1, 8), (2, 7), (3, 6), (4, 5). Four numbers can take one from each pair, for example 1, 2, 3, 4. With 5 numbers two must come from the same pair (pigeonhole).

ARYABHATTA MATHS OLYMPIAD

Level 2

PRACTICE PAPER-8

CLASS 11

Total Questions	30 multiple-choice questions
Level	Level-2 (final round): every question is Hard
Time	45 minutes
Arrangement	Parts by chapter; the last Part has mixed problems joining two chapters
Negative Marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Algebra · Geometry · Trigonometry · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning

Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

For teachers

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2, 3 and 6 are not part of the Class 11 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

Part 1 — Number System (Syllabus Chapter 1)

Q1. Let $z = (2 + 3i)(1 - i)$. What is the value of $z \cdot \bar{z}$ minus the real part of z^2?

(A) 26 **(B)** 24 **(C)** 2 **(D)** −22

Q2. Let $z = 2 - 3i$. What is the value of $z^3 - 3z^2 + 9z + 4$?

(A) 4 **(B)** −9 **(C)** 13 **(D)** $-9 + 12i$

Q3. If $|z| = 2$, what is the difference between the greatest and the least possible values of $|z + 3 - 4i|$?

(A) 4 **(B)** 10 **(C)** 2 **(D)** 7

Part 2 — Algebra (Syllabus Chapter 4)

Q4. How many numbers are terms of both the AP 3, 7, 11, … , 407 and the AP 2, 9, 16, … , 709?

(A) 13 **(B)** 15 **(C)** 28 **(D)** 14

Q5. What is the value of $1 \cdot 10 + 2 \cdot 9 + 3 \cdot 8 + \ldots + 10 \cdot 1$?

(A) 385 **(B)** 165 **(C)** 220 **(D)** 286

Q6. Three geometric means are inserted between 2 and 18. What is their product?

(A) 36 **(B)** 216 **(C)** 108 **(D)** $72\sqrt{3}$

Q7. What is the middle term of the expansion of $(x/2 + 2/x)^8$?

(A) 70 **(B)** 35 **(C)** 70x **(D)** 1120

Q8. Positive real numbers x, y, z satisfy $xyz = 8$. What is the least value of $x + 2y + 4z$?

(A) 7 **(B)** 6 **(C)** 14 **(D)** 12

Q9. How many integers x satisfy both $(x - 2)/(x + 3) \le 0$ and $|2x - 1| > 3$?

(A) 1 **(B)** 5 **(C)** 2 **(D)** 3

Part 3 — Geometry (Syllabus Chapter 5)

Q10. Where does the perpendicular bisector of the segment joining (1, 3) and (5, 7) meet the x-axis?

(A) (3, 0) **(B)** (−8, 0) **(C)** (8, 0) **(D)** (5, 0)

Q11. Of all the lines through the point where $x + y = 3$ meets $2x − y = 0$, which one is farthest from the origin?

(A) $2x − y = 0$ **(B)** $x + 2y = 5$ **(C)** $x + y = 3$ **(D)** $2x + y = 4$

Q12. What is the area of the triangle formed by the lines $2x − y = 0$, $x + 2y = 10$ and $y = 0$?

(A) 10 **(B)** 20 **(C)** 40 **(D)** 25

Part 4 — Trigonometry (Syllabus Chapter 7)

Q13. What is the value of $\cos 20° + \cos 100° + \cos 140°$?

(A) 1 **(B)** 1/2 **(C)** 0 **(D)** −1/2

Q14. For every θ where it is defined, $\sin 3\theta/\sin \theta − \cos 3\theta/\cos \theta$ equals which constant?

(A) 1 **(B)** 4 **(C)** 0 **(D)** 2

Q15. What is the value of $(1 + \tan 1°)(1 + \tan 2°)(1 + \tan 3°) \ldots (1 + \tan 44°)$?

(A) 2^{22} **(B)** 2^{44} **(C)** 2^{11} **(D)** 2^{23}

Part 5 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Q16. In how many ways can two squares of an 8×8 chessboard be chosen so that they are not in the same row or the same column?

(A) 2016 **(B)** 3136 **(C)** 784 **(D)** 1568

Q17. In how many arrangements of the letters of ARRANGE are the two R's together but the two A's not together?

(A) 360 **(B)** 120 **(C)** 240 **(D)** 480

Q18. Two dice are thrown together 3 times. What is the probability that a total of 7 appears at least once?

(A) 1/2 **(B)** 125/216 **(C)** 1/6 **(D)** 91/216

Q19. Team A scores have mean 50 and SD 5. Team B scores have mean 40 and SD 6. What is the coefficient of variation of team B, and which team is more consistent?

(A) 15%, team A **(B)** 15%, team B **(C)** 10%, team A **(D)** 6%, team B

Part 6 — Applied Mathematics (Syllabus Chapter 9)

Q20. A deposit earns 12% per annum compounded every quarter. What is the effective annual rate, to two decimal places?

(A) 12.36% **(B)** 12.00% **(C)** 12.55% **(D)** 12.68%

Q21. A 25-litre vessel is full of a 32% acid solution. Twice, x litres are taken out and replaced with water. The solution is now 20.48% acid. What is x?

(A) 4 L **(B)** 5 L **(C)** 8 L **(D)** 6 L

Q22. Plan A costs ₹200 a month plus ₹2 per minute. Plan B costs ₹500 a month plus ₹0.50 per minute. For which monthly usage is Plan B strictly cheaper?

(A) More than 150 minutes **(B)** More than 200 minutes

(C) Less than 200 minutes **(D)** More than 300 minutes

Part 7 — Mathematical Reasoning (Syllabus Chapter 10)

Q23. What is the largest number of bishops that can be placed on an 8 × 8 chessboard so that no two attack each other?

(A) 14 **(B)** 8 **(C)** 16 **(D)** 15

Q24. What is the least number of points with integer coordinates that guarantees two of them have a midpoint with integer coordinates?

(A) 4 **(B)** 9 **(C)** 3 **(D)** 5

Q25. What is the largest number of regions into which 5 circles can divide the plane?

(A) 32 **(B)** 20 **(C)** 22 **(D)** 11

Part 8 — Mixed Problems (each joins two chapters)

Q26. What is the value of $1/(1\cdot2\cdot3) + 1/(2\cdot3\cdot4) + \ldots + 1/(10\cdot11\cdot12)$?

(A) 1/4 **(B)** 65/264 **(C)** 55/264 **(D)** 1/3

Q27. What is the value of $C(8, 0)/1 + C(8, 1)/2 + C(8, 2)/3 + \ldots + C(8, 8)/9$?

(A) 256/9 **(B)** 511/9 **(C)** 512/9 **(D)** 255/8

Q28. What is the smallest positive integer n for which $i \times i^2 \times i^3 \times \ldots \times i^n = 1$?

(A) 7 **(B)** 3 **(C)** 4 **(D)** 8

Q29. Which line through the origin bisects the acute angle between the x-axis and the line $y = \sqrt{3}x$?

(A) $y = x$ **(B)** $y = \sqrt{3}x/2$ **(C)** $y = -\sqrt{3}x$ **(D)** $y = x/\sqrt{3}$

Q30. Three of the 9 points of a 3 × 3 square grid of dots are chosen at random. What is the probability that they lie on a straight line?

(A) 1/12 **(B)** 1/14 **(C)** 8/27 **(D)** 2/21

Answer Sheet AMO Level 2 · Class 11 · Practice Paper 8

Name: **School:** **Roll No.:**

Shade or tick one circle for each question.

1.	A	B	C	D	11.	A	B	C	D	21.	A	B	C	D
2.	A	B	C	D	12.	A	B	C	D	22.	A	B	C	D
3.	A	B	C	D	13.	A	B	C	D	23.	A	B	C	D
4.	A	B	C	D	14.	A	B	C	D	24.	A	B	C	D
5.	A	B	C	D	15.	A	B	C	D	25.	A	B	C	D
6.	A	B	C	D	16.	A	B	C	D	26.	A	B	C	D
7.	A	B	C	D	17.	A	B	C	D	27.	A	B	C	D
8.	A	B	C	D	18.	A	B	C	D	28.	A	B	C	D
9.	A	B	C	D	19.	A	B	C	D	29.	A	B	C	D
10.	A	B	C	D	20.	A	B	C	D	30.	A	B	C	D

For the parent or teacher

Correct: _____ / 30

Answer Key and Solutions AMO Level 2 · Class 11 · Practice Paper 8

Quick answers

1 **C**	6 **B**	11 **B**	16 **D**	21 **B**	26 **B**
2 **B**	7 **A**	12 **B**	17 **C**	22 **B**	27 **B**
3 **A**	8 **D**	13 **C**	18 **D**	23 **A**	28 **A**
4 **D**	9 **A**	14 **D**	19 **A**	24 **D**	29 **D**
5 **C**	10 **C**	15 **A**	20 **C**	25 **C**	30 **D**

Solutions

Q	Part · level	Answer	Why?
1	Part 1 · Ch1 · Hard	**(C)** 2	***Hint:*** *Multiply out z first, then use $z\cdot\bar{z} = \|z\|^2$.* $z = 2 - 2i + 3i - 3i^2 = 5 + i$. Then $z\cdot\bar{z} = 5^2 + 1^2 = 26$ and $z^2 = 25 + 10i - 1 = 24 + 10i$, so its real part is 24. The answer is $26 - 24 = 2$.
2	Part 1 · Ch1 · Hard	**(B)** −9	***Hint:*** *z satisfies $z^2 - 4z + 13 = 0$. Divide the cubic by this quadratic.* $(z - 2)^2 = -9$ gives $z^2 - 4z + 13 = 0$. $z^3 - 3z^2 + 9z + 4 = (z + 1)(z^2 - 4z + 13) - 9$, so the value is −9.
3	Part 1 · Ch1 · Hard	**(A)** 4	***Hint:*** *Use the triangle inequality both ways with $\|3 - 4i\| = 5$.* $\|z + 3 - 4i\| \le \|z\| + \|3 - 4i\| = 7$ and $\ge \|3 - 4i\| - \|z\| = 3$. Both values are reached ($z = \pm(2/5)(3 - 4i)$). Difference $7 - 3 = 4$.
4	Part 2 · Ch4 · Hard	**(D)** 14	***Hint:*** *The common terms form an AP whose difference is the LCM of 4 and 7.* The first common term is 23 ($23 = 3 + 5 \times 4 = 2 + 3 \times 7$). Common terms: 23, 51, 79, … with difference 28. The last one not above 407 is $23 + 13 \times 28 = 387$. That is 14 terms.
5	Part 2 · Ch4 · Hard	**(C)** 220	***Hint:*** *The kth term is $k(11 - k) = 11k - k^2$.* $\Sigma(11k - k^2)$ for $k = 1$ to $10 = 11 \times 55 - 385 = 605 - 385 = 220$.
6	Part 2 · Ch4 · Hard	**(B)** 216	***Hint:*** *Equally spaced terms of a GP pair up to the same product.* $2, G_1, G_2, G_3, 18$ is a GP. $G_1G_3 = G_2^2 = 2 \times 18 = 36$, so $G_2 = 6$. Product $= 36 \times 6 = 216$. (The means are $2\sqrt{3}$, 6, $6\sqrt{3}$.)
7	Part 2 · Ch4 · Hard	**(A)** 70	***Hint:*** *With 9 terms, the middle one is the 5th.* $T_5 = C(8, 4)(x/2)^4(2/x)^4 = 70 \times (x^4/16) \times (16/x^4) = 70$.
8	Part 2 · Ch4 · Hard	**(D)** 12	***Hint:*** *Apply AM ≥ GM to x, $2y$ and $4z$.* $(x + 2y + 4z)/3 \ge \sqrt[3]{(x \times 2y \times 4z)} = \sqrt[3]{64} = 4$. So $x + 2y + 4z \ge 12$, with equality at $x = 4$, $y = 2$, $z = 1$.
9	Part 2 · Ch4 · Hard	**(A)** 1	***Hint:*** *Solve each inequality separately, then take the common integers.* The first gives $-3 < x \le 2$. The second gives $2x - 1 > 3$ or $2x - 1 < -3$, that is $x > 2$ or $x < -1$. Common integers: only $x = -2$. So 1.
10	Part 3 · Ch5 · Hard	**(C)** (8, 0)	***Hint:*** *It passes through the midpoint and is perpendicular to the segment.* Midpoint (3, 5). The segment has slope 1, so the bisector has slope −1: $y - 5 = -(x - 3)$, that is $x + y = 8$. It meets the x-axis at (8, 0).

Q	Part · level	Answer	Why?
11	Part 3 · Ch5 · Hard	**(B)** x + 2y = 5	***Hint:*** *The farthest line is perpendicular to the segment joining the origin to the meeting point.* Adding the equations: 3x = 3, so the point is P(1, 2). The distance from O is greatest (= OP = √5) when the line is perpendicular to OP, whose slope is 2. So the slope is −1/2: y − 2 = −½(x − 1), that is x + 2y = 5.
12	Part 3 · Ch5 · Hard	**(B)** 20	***Hint:*** *The first two lines are perpendicular. Find all three corners.* Corners: (0, 0), (10, 0) and the meeting point of 2x − y = 0 and x + 2y = 10, which is (2, 4). Area = ½ × 10 × 4 = 20.
13	Part 4 · Ch7 · Hard	**(C)** 0	***Hint:*** *Combine cos 100° + cos 140° with sum-to-product.* cos 100° + cos 140° = 2 cos 120° cos 20° = −cos 20°. So the total is cos 20° − cos 20° = 0.
14	Part 4 · Ch7 · Hard	**(D)** 2	***Hint:*** *Combine into one fraction: the top is sin(3θ − θ).* (sin 3θ cos θ − cos 3θ sin θ)/(sin θ cos θ) = sin 2θ/(sin θ cos θ) = 2 sin θ cos θ/(sin θ cos θ) = 2.
15	Part 4 · Ch7 · Hard	**(A)** 2^{22}	***Hint:*** *If A + B = 45°, then (1 + tan A)(1 + tan B) = 2.* tan(A + B) = 1 gives tan A + tan B = 1 − tan A tan B, so (1 + tan A)(1 + tan B) = 2. Pair 1° with 44°, 2° with 43°, …, 22° with 23°: 22 pairs. The product is 2^{22}.
16	Part 5 · Ch8 · Hard	**(D)** 1568	***Hint:*** *Choose the first square, then a square outside its row and column, and divide by 2 for order.* 64 choices for one square and 49 for the other (outside its row and column): 64 × 49/2 = 1568. (Equivalently C(64, 2) − 448 = 1568.)
17	Part 5 · Ch8 · Hard	**(C)** 240	***Hint:*** *Treat RR as one letter. Then subtract the arrangements where AA is also together.* With RR as one block: letters A, A, RR, N, G, E give 6!/2! = 360. With AA also a block: 5! = 120. Answer 360 − 120 = 240.
18	Part 5 · Ch8 · Hard	**(D)** 91/216	***Hint:*** *A total of 7 has probability 1/6 on each throw. Use the complement.* P(no 7 in 3 throws) = $(5/6)^3$ = 125/216. P(at least one 7) = 1 − 125/216 = 91/216.
19	Part 5 · Ch8 · Hard	**(A)** 15%, team A	***Hint:*** *Coefficient of variation = SD/mean × 100%. Smaller means more consistent.* CV(B) = 6/40 × 100% = 15%. CV(A) = 5/50 × 100% = 10%. Team A has the smaller CV, so it is more consistent.
20	Part 6 · Ch9 · Hard	**(C)** 12.55%	***Hint:*** *Each quarter multiplies the money by 1.03.* 1.03^4 = 1.1255…, so the effective rate is 12.55%. (12.36% is half-yearly compounding.)
21	Part 6 · Ch9 · Hard	**(B)** 5 L	***Hint:*** *Each replacement multiplies the acid fraction by (1 − x/25).* 0.32 × $(1 − x/25)^2$ = 0.2048, so $(1 − x/25)^2$ = 0.64 and 1 − x/25 = 0.8. x = 5 L.
22	Part 6 · Ch9 · Hard	**(B)** More than 200 minutes	***Hint:*** *Solve 500 + 0.5m < 200 + 2m.* 500 + 0.5m < 200 + 2m gives 300 < 1.5m, so m > 200. Plan B is cheaper for more than 200 minutes.

Q	Part · level	Answer	Why?
23	Part 7 · Ch10 · Hard	**(A)** 14	***Hint:*** *Look at the 15 diagonals in one direction. Two of them have only one square, at opposite corners.* There are 15 diagonals running one way, and each can hold at most one bishop. The two one-square diagonals are opposite corners, which lie on the same diagonal the other way, so they cannot both be used: at most 14. Placing 8 bishops on the first row and 6 on the last row (not in the corners) gives 14.
24	Part 7 · Ch10 · Hard	**(D)** 5	***Hint:*** *Sort the points by the parity of their coordinates.* There are 4 parity classes: (even, even), (even, odd), (odd, even), (odd, odd). With 5 points, two share a class, and then both coordinates of their midpoint are integers. The four points (0, 0), (0, 1), (1, 0), (1, 1) show 4 is not enough.
25	Part 7 · Ch10 · Hard	**(C)** 22	***Hint:*** *The kth circle can meet each earlier circle in 2 points.* The kth circle meets the earlier k − 1 circles in at most 2(k − 1) points, cutting itself into 2(k − 1) arcs, each adding a region. Starting with 2 regions for one circle: 2 + 2 + 4 + 6 + 8 = 22 (formula $n^2 - n + 2$).
26	Part 8 · Mixed · Ch4 + Ch10 · Hard	**(B)** 65/264	***Hint:*** *$1/(k(k+1)(k+2)) = \frac{1}{2}[1/(k(k+1)) - 1/((k+1)(k+2))]$. Prove the formula by induction if you like.* The sum telescopes to $\frac{1}{2}[1/(1\cdot2) - 1/(11\cdot12)] = \frac{1}{2}(1/2 - 1/132) = \frac{1}{2} \times 65/132 = 65/264$.
27	Part 8 · Mixed · Ch4 + Ch8 · Hard	**(B)** 511/9	***Hint:*** *$C(8, k)/(k + 1) = C(9, k + 1)/9$.* $\Sigma\, C(8, k)/(k + 1) = (1/9)\, \Sigma\, C(9, k + 1) = (1/9)(2^9 - 1) = 511/9$.
28	Part 8 · Mixed · Ch1 + Ch4 · Hard	**(A)** 7	***Hint:*** *The product is i raised to $1 + 2 + \ldots + n = n(n + 1)/2$.* We need $n(n + 1)/2$ to be a multiple of 4. n = 1, 2, ..., 6 give 1, 3, 6, 10, 15, 21, none a multiple of 4. n = 7 gives 28. So n = 7.
29	Part 8 · Mixed · Ch5 + Ch7 · Hard	**(D)** $y = x/\sqrt{3}$	***Hint:*** *The line $y = \sqrt{3}x$ makes 60° with the x-axis.* The acute angle is from 0° to 60°, so the bisector makes 30° with the x-axis: slope $\tan 30° = 1/\sqrt{3}$, giving $y = x/\sqrt{3}$.
30	Part 8 · Mixed · Ch8 + Ch10 · Hard	**(D)** 2/21	***Hint:*** *Count the lines of three dots: rows, columns and diagonals.* There are 3 rows, 3 columns and 2 diagonals: 8 collinear triples. Total choices C(9, 3) = 84. P = 8/84 = 2/21.

ARYABHATTA MATHS OLYMPIAD

Level 2

PRACTICE PAPER-9

CLASS 11

Total Questions	30 multiple-choice questions
Level	Level-2 (final round): every question is Hard
Time	45 minutes
Arrangement	Parts by chapter; the last Part has mixed problems joining two chapters
Negative Marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Algebra · Geometry · Trigonometry · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning

Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

For teachers

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2, 3 and 6 are not part of the Class 11 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

Part 1 — Number System (Syllabus Chapter 1)

Q1. For how many real k is $(1 + ki)^3$ purely imaginary, and what is the product of these values of k?

(A) 2 values, product 1/3 **(B)** 1 value, k = 0

(C) 3 values, product 0 **(D)** 2 values, product −1/3

Q2. A complex number z satisfies $|z - 4/z| = 2$. What is the greatest possible value of $|z|$?

(A) $\sqrt{5}$ **(B)** $1 + \sqrt{5}$ **(C)** 2 **(D)** 3

Q3. For how many integers n from 1 to 50 is $i^n + (-i)^n = 2$?

(A) 25 **(B)** 13 **(C)** 12 **(D)** 24

Part 2 — Algebra (Syllabus Chapter 4)

Q4. In an AP the 5th term is 1/8 and the 8th term is 1/5. What is its 40th term?

(A) 1 **(B)** 1/40 **(C)** 40 **(D)** 9/40

Q5. Which formula gives the sum of the first n terms of 7 + 77 + 777 + … ?

(A) $(7/9)(10^n - 1)$ **(B)** $(7/81)(10^{n+1} - 9n - 10)$

(C) $(7/81)(10^n - 9n - 1)$ **(D)** $(7/9)(10^{n+1} - 10)$

Q6. What is the value of $11^2 + 12^2 + 13^2 + \ldots + 20^2$?

(A) 2485 **(B)** 2870 **(C)** 2585 **(D)** 2385

Q7. In the expansion of $(1 + x)^n$, the sum of the coefficients of the odd powers of x is 512. What is the coefficient of x^3?

(A) 120 **(B)** 84 **(C)** 165 **(D)** 45

Q8. The arithmetic mean and the geometric mean of two positive numbers are in the ratio 5 : 4. What is the ratio of the larger number to the smaller?

(A) 5 : 4 **(B)** 25 : 16 **(C)** 4 : 1 **(D)** 9 : 1

Q9. How many points (x, y) with x and y positive integers satisfy x + y < 10?

(A) 45 **(B)** 36 **(C)** 28 **(D)** 81

Part 3 — Geometry (Syllabus Chapter 5)

Q10. What is the reflection (mirror image) of the point (2, 3) in the line 3x − 4y = 0?

(A) (86/25, 27/25) **(B)** (68/25, 51/25)

(C) (3, 2) **(D)** (−2, −3)

Q11. For every real k, the line (2 + k)x + (3 − k)y = 5 passes through one fixed point. Which point?

(A) (2, 3) **(B)** (1, 1) **(C)** (5, 0) **(D)** (1, −1)

Q12. θ is the acute angle between the lines 2x − y + 3 = 0 and x + 3y − 1 = 0. What is tan θ?

(A) 1/7 **(B)** 7 **(C)** 1 **(D)** 5/7

Part 4 — Trigonometry (Syllabus Chapter 7)

Q13. Simplify sin 20° + sin 40° − sin 80° completely.

(A) 1/2 **(B)** √3/2 **(C)** 0 **(D)** −1/2

Q14. Angle A lies between 270° and 360°, and cos A = 3/5. What is tan(A/2)?

(A) 1/2 **(B)** −2 **(C)** 2 **(D)** −1/2

Q15. What is the value of cot 15° − tan 15°?

(A) 2√3 **(B)** 4 **(C)** 2 **(D)** √3

Part 5 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Q16. How many positive integers less than 1000 have digits that add up to 10?

(A) 66 **(B)** 55 **(C)** 54 **(D)** 63

Q17. Seven people stand in a row. Three particular friends must stand together, and two other particular people, Dev and Esha, must not stand next to each other. In how many ways can they stand?

(A) 720 **(B)** 432 **(C)** 288 **(D)** 480

Q18. In how many different ways can 7 different beads be strung on a necklace? (A necklace can be turned over.)

(A) 720 **(B)** 5040 **(C)** 2520 **(D)** 360

Q19. A fair die is thrown 30 times. X is the number of sixes. What is the standard deviation of X?

(A) 25/6 **(B)** 5 **(C)** √5 **(D)** 5/√6

Part 6 — Applied Mathematics (Syllabus Chapter 9)

Q20. Sahil can do a job in 12 days and Tanvi in 15 days. They work together for 3 days, then Sahil leaves and Tanvi finishes the job. How many days does the whole job take?

(A) 11¼ days **(B)** 8¼ days **(C)** 9 days **(D)** 12 days

Q21. A trader buys goods from another state for ₹10,000 plus 18% IGST and sells them in her own state for ₹15,000 plus 18% GST (CGST + SGST). After input tax credit, how much tax does she pay?

(A) ₹2,700 **(B)** ₹900 **(C)** ₹1,800 **(D)** ₹450

Q22. A boat goes 24 km downstream in 2 hours and comes back the same 24 km upstream in 3 hours. If the stream then flows twice as fast, how long will the boat take to go 30 km upstream?

(A) 3.75 h **(B)** 7.5 h **(C)** 5 h **(D)** 6 h

Part 7 — Mathematical Reasoning (Syllabus Chapter 10)

Q23. What is the largest number that divides $3^{2n} - 1$ for every natural number n?

(A) 8 **(B)** 4 **(C)** 16 **(D)** 24

Q24. A false proof claims that in any group of n horses all have the same colour. The step takes a group of k + 1 horses, removes one horse, then another, and uses two overlapping groups of k. For which step does this argument fail?

(A) k = 2 to 3 **(B)** The base case n = 1

(C) k = 1 to 2 **(D)** Every step from k = 5

Q25. How many squares of all sizes (formed by the grid lines) are there on an 8 × 8 chessboard?

(A) 64 **(B)** 1296 **(C)** 120 **(D)** 204

Part 8 — Mixed Problems (each joins two chapters)

Q26. Four rings of different sizes are stacked on the left peg of three pegs in a row, largest at the bottom. A move takes the top ring of one peg to another peg, never onto a smaller ring, and every move must go to or from the middle peg. What is the least number of moves to shift the stack to the right peg?

(A) 15 **(B)** 26 **(C)** 80 **(D)** 81

Q27. How many different terms are there in the expansion of $(x + y + z)^4$, after collecting like terms?

(A) 12 **(B)** 81 **(C)** 15 **(D)** 10

Q28. What is the value of $C(6, 0) + C(6, 1)i + C(6, 2)i^2 + \ldots + C(6, 6)i^6$?

(A) 8i **(B)** −8i **(C)** −8 **(D)** 64

Q29. Two lines through the origin each make an angle of 60° with the y-axis. What is the product of their slopes?

(A) −3 **(B)** 1/3 **(C)** −1 **(D)** −1/3

Q30. A path goes from (0, 0) to (4, 4) in unit steps right or up, and never goes above the line y = x. How many such paths are there?

(A) 70 **(B)** 35 **(C)** 14 **(D)** 42

Answer Sheet AMO Level 2 · Class 11 · Practice Paper 9

Name: **School:** **Roll No.:**

Shade or tick one circle for each question.

1.	(A)	(B)	(C)	(D)	11.	(A)	(B)	(C)	(D)	21.	(A)	(B)	(C)	(D)
2.	(A)	(B)	(C)	(D)	12.	(A)	(B)	(C)	(D)	22.	(A)	(B)	(C)	(D)
3.	(A)	(B)	(C)	(D)	13.	(A)	(B)	(C)	(D)	23.	(A)	(B)	(C)	(D)
4.	(A)	(B)	(C)	(D)	14.	(A)	(B)	(C)	(D)	24.	(A)	(B)	(C)	(D)
5.	(A)	(B)	(C)	(D)	15.	(A)	(B)	(C)	(D)	25.	(A)	(B)	(C)	(D)
6.	(A)	(B)	(C)	(D)	16.	(A)	(B)	(C)	(D)	26.	(A)	(B)	(C)	(D)
7.	(A)	(B)	(C)	(D)	17.	(A)	(B)	(C)	(D)	27.	(A)	(B)	(C)	(D)
8.	(A)	(B)	(C)	(D)	18.	(A)	(B)	(C)	(D)	28.	(A)	(B)	(C)	(D)
9.	(A)	(B)	(C)	(D)	19.	(A)	(B)	(C)	(D)	29.	(A)	(B)	(C)	(D)
10.	(A)	(B)	(C)	(D)	20.	(A)	(B)	(C)	(D)	30.	(A)	(B)	(C)	(D)

For the parent or teacher

Correct: _____ / 30

Answer Key and Solutions AMO Level 2 · Class 11 · Practice Paper 9

Quick answers

1 **D**	6 **A**	11 **B**	16 **D**	21 **B**	26 **C**
2 **B**	7 **A**	12 **B**	17 **B**	22 **C**	27 **C**
3 **C**	8 **C**	13 **C**	18 **D**	23 **A**	28 **B**
4 **A**	9 **B**	14 **D**	19 **D**	24 **C**	29 **D**
5 **B**	10 **A**	15 **A**	20 **A**	25 **D**	30 **C**

Solutions

Q	Part · level	Answer	Why?
1	Part 1 · Ch1 · Hard	**(D)** 2 values, product −1/3	***Hint:*** *Expand $(1 + ki)^3$ and set the real part to 0.* $(1 + ki)^3 = 1 + 3ki - 3k^2 - k^3 i$. Real part $1 - 3k^2 = 0$ gives $k = \pm 1/\sqrt{3}$. For these, the imaginary part $3k - k^3 \neq 0$. Two values with product −1/3.
2	Part 1 · Ch1 · Hard	**(B)** $1 + \sqrt{5}$	***Hint:*** *Use $\lvert z\rvert - 4/\lvert z\rvert \le \lvert z - 4/z\rvert$.* Let $r = \lvert z\rvert$. Then $r - 4/r \le 2$, so $r^2 - 2r - 4 \le 0$ and $r \le 1 + \sqrt{5}$. It is reached at $z = 1 + \sqrt{5}$: then $4/z = \sqrt{5} - 1$, so $z - 4/z = 2$. The greatest value is $1 + \sqrt{5}$.
3	Part 1 · Ch1 · Hard	**(C)** 12	***Hint:*** *$(-i)^n = (-1)^n i^n$. Consider odd and even n.* $i^n + (-i)^n = i^n(1 + (-1)^n)$. For odd n this is 0. For even n it is $2i^n$, which equals 2 when $i^n = 1$, that is n is a multiple of 4. n = 4, 8, … , 48: 12 values.
4	Part 2 · Ch4 · Hard	**(A)** 1	***Hint:*** *Find d from the two given terms, then the first term.* $3d = 1/5 - 1/8 = 3/40$, so $d = 1/40$. $a = 1/8 - 4/40 = 1/40$. So $T_n = n/40$ and $T_{40} = 1$.
5	Part 2 · Ch4 · Hard	**(B)** $(7/81)(10^{n+1} - 9n - 10)$	***Hint:*** *The kth term is $(7/9)(10^k - 1)$. Check your formula at n = 1.* Sum $= (7/9)[(10 + 100 + \ldots + 10^n) - n] = (7/9)[10(10^n - 1)/9 - n] = (7/81)(10^{n+1} - 10 - 9n)$. Check: n = 1 gives (7/81)(81) = 7.
6	Part 2 · Ch4 · Hard	**(A)** 2485	***Hint:*** *Subtract Σk^2 up to 10 from Σk^2 up to 20.* $\Sigma_1^{20} k^2 = 20 \cdot 21 \cdot 41/6 = 2870$. $\Sigma_1^{10} k^2 = 385$. Difference 2485.
7	Part 2 · Ch4 · Hard	**(A)** 120	***Hint:*** *The sum of the odd-power coefficients is 2^{n-1}.* $2^{n-1} = 512 = 2^9$, so n = 10. The coefficient of x^3 is C(10, 3) = 120.
8	Part 2 · Ch4 · Hard	**(C)** 4 : 1	***Hint:*** *Take AM = 5k and GM = 4k, so the numbers have sum 10k and product $16k^2$.* The numbers are the roots of $t^2 - 10kt + 16k^2 = 0$, that is t = 8k and t = 2k. Ratio 8k : 2k = 4 : 1.
9	Part 2 · Ch4 · Hard	**(B)** 36	***Hint:*** *For x + y = s, there are s − 1 points. s runs from 2 to 9.* Sum over s = 2 to 9 of (s − 1) = 1 + 2 + … + 8 = 36.
10	Part 3 · Ch5 · Hard	**(A)** (86/25, 27/25)	***Hint:*** *Move along the normal direction (3, −4) twice the distance to the foot of the perpendicular.* $3(2) - 4(3) = -6$. Image $= (2, 3) - 2 \times (-6)/25 \times (3, -4) = (2 + 36/25, 3 - 48/25) = (86/25, 27/25)$. (68/25, 51/25) is the foot of the perpendicular.
11	Part 3 · Ch5 · Hard	**(B)** (1, 1)	***Hint:*** *Write the equation as $(2x + 3y - 5) + k(x - y) = 0$.* The equation holds for all k exactly when $2x + 3y - 5 = 0$ and $x - y = 0$. Then x = y and 5x = 5: the point (1, 1).

Q	Part · level	Answer	Why?
12	Part 3 · Ch5 · Hard	**(B)** 7	***Hint:*** *The slopes are 2 and −1/3.* tan θ = |(2 − (−1/3))/(1 + 2(−1/3))| = (7/3)/(1/3) = 7.
13	Part 4 · Ch7 · Hard	**(C)** 0	***Hint:*** *sin 20° + sin 40° = 2 sin 30° cos 10°.* sin 20° + sin 40° = 2 sin 30° cos 10° = cos 10° = sin 80°. So the value is 0.
14	Part 4 · Ch7 · Hard	**(D)** −1/2	***Hint:*** *Use tan(A/2) = sin A/(1 + cos A), and decide the quadrant of A/2.* sin A = −4/5. tan(A/2) = (−4/5)/(1 + 3/5) = −1/2. Check: A/2 lies between 135° and 180°, where tan is negative.
15	Part 4 · Ch7 · Hard	**(A)** 2√3	***Hint:*** *cot θ − tan θ = 2 cot 2θ.* cot θ − tan θ = (cos²θ − sin²θ)/(sin θ cos θ) = 2 cos 2θ/sin 2θ = 2 cot 2θ. At 15°: 2 cot 30° = 2√3.
16	Part 5 · Ch8 · Hard	**(D)** 63	***Hint:*** *Count solutions of a + b + c = 10 with each digit from 0 to 9.* Non-negative solutions of a + b + c = 10: C(12, 2) = 66. Remove those with a digit equal to 10 (the others 0): 3. Answer 63.
17	Part 5 · Ch8 · Hard	**(B)** 432	***Hint:*** *Count the arrangements with the friends together, then subtract those where Dev and Esha are also together.* Friends together: 5! × 3! = 720. Of these, Dev and Esha together: 4! × 2! × 3! = 288. Answer: 720 − 288 = 432.
18	Part 5 · Ch8 · Hard	**(D)** 360	***Hint:*** *Round arrangements of n items: (n − 1)!. Turning over pairs them up.* Circular arrangements: 6! = 720. Turning the necklace over makes each arrangement the same as its mirror image, so divide by 2: 360.
19	Part 5 · Ch8 · Hard	**(D)** 5/√6	***Hint:*** *For a binomial variable, variance = npq.* n = 30, p = 1/6, q = 5/6. Variance = 30 × 1/6 × 5/6 = 25/6. SD = 5/√6. (25/6 is the variance and 5 the mean.)
20	Part 6 · Ch9 · Hard	**(A)** 11¼ days	***Hint:*** *Find the part done in the first 3 days.* Together: 1/12 + 1/15 = 9/60 a day, so 27/60 in 3 days. Left: 33/60. Tanvi takes 33/60 × 15 = 8¼ days. Total 3 + 8¼ = 11¼ days. (8¼ is only Tanvi's part.)
21	Part 6 · Ch9 · Hard	**(B)** ₹900	***Hint:*** *Tax collected on the sale minus tax paid on the purchase.* Tax collected: 18% of 15,000 = ₹2,700 (₹1,350 CGST + ₹1,350 SGST). Credit for IGST paid: ₹1,800. Net payable: ₹900.
22	Part 6 · Ch9 · Hard	**(C)** 5 h	***Hint:*** *Find the boat speed and the stream speed first.* Downstream 12 km/h, upstream 8 km/h, so the boat is 10 km/h and the stream 2 km/h. With the stream at 4 km/h, upstream speed is 6 km/h: 30 ÷ 6 = 5 h.
23	Part 7 · Ch10 · Hard	**(A)** 8	***Hint:*** *Check n = 1, then prove divisibility by that number by induction.* n = 1 gives 8, so the answer is at most 8. Induction: $3^{2(k+1)} - 1 = 9(3^{2k} - 1) + 8$, which is a multiple of 8 if $3^{2k} - 1$ is. So 8 works for all n.
24	Part 7 · Ch10 · Hard	**(C)** k = 1 to 2	***Hint:*** *The two smaller groups must share at least one horse.* For k + 1 = 2 horses, removing each horse in turn leaves two groups of one horse with no horse in common, so nothing links their colours. For k ≥ 2 the groups overlap and the step is valid. The base case n = 1 is true.
25	Part 7 · Ch10 · Hard	**(D)** 204	***Hint:*** *A k × k square can be placed in (9 − k)² positions.* Σ (9 − k)² for k = 1 to 8 = 8² + 7² + … + 1² = 204. (1296 counts all rectangles.)

Q	Part · level	Answer	Why?
26	Part 8 · Mixed · Ch4 + Ch10 · Hard	**(C)** 80	***Hint:*** *Let a_n be the moves for n rings. The largest ring must go left → middle → right.* To move the largest ring twice, the other n − 1 rings must go right, back left, and right again: $a_n = 3a_{n-1} + 2$ with $a_1 = 2$. So $a_n = 3^n - 1$, and $a_4 = 80$. (15 is the ordinary puzzle.)
27	Part 8 · Mixed · Ch4 + Ch8 · Hard	**(C)** 15	***Hint:*** *Each term is $x^a y^b z^c$ with $a + b + c = 4$.* The number of non-negative solutions of a + b + c = 4 is C(6, 2) = 15. ($81 = 3^4$ counts the products before collecting.)
28	Part 8 · Mixed · Ch1 + Ch4 · Hard	**(B)** −8i	***Hint:*** *It is the binomial expansion of $(1 + i)^6$.* The sum is $(1 + i)^6 = (2i)^3 = 8i^3 = -8i$.
29	Part 8 · Mixed · Ch5 + Ch7 · Hard	**(D)** −1/3	***Hint:*** *A line making 60° with the y-axis makes 30° with the x-axis.* The lines make 30° and 150° with the positive x-axis, so their slopes are tan 30° = 1/√3 and tan 150° = −1/√3. Product = −1/3.
30	Part 8 · Mixed · Ch8 + Ch10 · Hard	**(C)** 14	***Hint:*** *Count the bad paths by reflecting the part after they first touch y = x + 1.* All paths: C(8, 4) = 70. Bad paths touch y = x + 1; reflecting the part after the first touch in y = x + 1 gives a path to (3, 5), and there are C(8, 3) = 56 of them. Good paths: 70 − 56 = 14.

ARYABHATTA MATHS OLYMPIAD

Level 2

PRACTICE PAPER-10

CLASS 11

Total Questions	30 multiple-choice questions
Level	Level-2 (final round): every question is Hard
Time	45 minutes
Arrangement	Parts by chapter; the last Part has mixed problems joining two chapters
Negative Marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Algebra · Geometry · Trigonometry · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning

Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

For teachers

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2, 3 and 6 are not part of the Class 11 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

Part 1 — Number System (Syllabus Chapter 1)

Q1. What is the principal argument of $(1 - i)^3$ (the value in the range $-180° < \theta \le 180°$)?

(A) $-135°$ **(B)** $135°$ **(C)** $-45°$ **(D)** $225°$

Q2. Real numbers x and y satisfy $x(2 + i) + y(1 - i)^2 = 6 - 5i$. What is xy?

(A) -12 **(B)** 6 **(C)** 8 **(D)** 12

Q3. What is the value of $(1 + i)^{11} + (1 - i)^{11}$?

(A) 64 **(B)** 0 **(C)** -64 **(D)** -32

Part 2 — Algebra (Syllabus Chapter 4)

Q4. What is the least number of terms of the GP 3, 6, 12, … whose sum is more than 3000?

(A) 10 **(B)** 9 **(C)** 11 **(D)** 1000

Q5. What is the value of $1 + (1 + 2) + (1 + 2 + 3) + \ldots + (1 + 2 + \ldots + 20)$?

(A) 1540 **(B)** 1330 **(C)** 210 **(D)** 1771

Q6. The coefficients of the 2nd, 3rd and 4th terms in the expansion of $(1 + x)^n$ are in AP. What is n?

(A) 6 **(B)** 7 **(C)** 8 **(D)** 5

Q7. What is the least value of $(x^2 - x + 1)/(x^2 + x + 1)$ for real x?

(A) 0 **(B)** 1 **(C)** 1/3 **(D)** -1

Q8. How many points (x, y) with x and y non-negative integers satisfy $2x + y \ge 4$, $x + y \le 5$ and $x \le 3$?

(A) 15 **(B)** 12 **(C)** 10 **(D)** 18

Q9. Three numbers a, b, c are in GP. Their sum is 26 and the sum of their squares is 364. What is b?

(A) 8 **(B)** 6 **(C)** 2 **(D)** 4

Part 3 — Geometry (Syllabus Chapter 5)

Q10. The line $2x - y = 3$ is reflected in the line $x = 1$. What is the equation of the image line?

(A) $2x + y + 3 = 0$ **(B)** $2x - y = 1$

(C) $2x + y = 3$ **(D)** $2x + y = 1$

Q11. The triangle with vertices (1, 1), (k, 3) and (5, 7) has area 4. What is the sum of all possible values of k?

(A) 11/3 **(B)** 1 **(C)** 14/3 **(D)** 7/3

Q12. A line through the point (2, 3) cuts the positive x-axis at A and the positive y-axis at B. What is the least possible area of triangle OAB?

(A) 6 **(B)** 24 **(C)** 12 **(D)** 10

Part 4 — Trigonometry (Syllabus Chapter 7)

Q13. What is the greatest value of $3\cos x + 4\sin x + 2$ over all angles x?

(A) 7 **(B)** 9 **(C)** 5 **(D)** 11

Q14. What is the value of $\sin^4 15° + \cos^4 15°$?

(A) 1 **(B)** 3/4 **(C)** 7/8 **(D)** 1/2

Q15. What is the exact value of $\tan 22.5°$?

(A) $\sqrt{2} + 1$ **(B)** $\sqrt{2} - 1$ **(C)** $1/\sqrt{2}$ **(D)** $2 - \sqrt{3}$

Part 5 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Q16. A team of 4 is to be chosen from 6 married couples so that no couple is in the team. In how many ways can it be done?

(A) 495 **(B)** 15 **(C)** 360 **(D)** 240

Q17. Five observations have mean 6 and variance 4. Three of them are 4, 6 and 8. What is the product of the other two observations?

(A) 36 **(B)** 32 **(C)** 30 **(D)** 24

Q18. Each of 4 independent shots hits the target with probability 2/3. What is the probability of at least 3 hits?

(A) 32/81 **(B)** 16/81 **(C)** 8/27 **(D)** 16/27

Q19. Ten values of x have $\Sigma x = 60$ and $\Sigma x^2 = 400$. What is the standard deviation of the values $2x + 3$?

(A) 2 **(B)** 7 **(C)** 16 **(D)** 4

Part 6 — Applied Mathematics (Syllabus Chapter 9)

Q20. A contractor can rent a machine for ₹500 a day, or buy it for ₹30,000 and pay ₹100 a day to run it. For how many days of use is buying cheaper?

(A) More than 60 days **(B)** More than 75 days

(C) More than 300 days **(D)** Less than 75 days

Q21. Kiran and Leena share profits in the ratio 3 : 2, but first Kiran, the working partner, takes 10% of the profit as salary. The profit is ₹50,000. How much does Kiran receive in all?

(A) ₹32,000 **(B)** ₹30,000 **(C)** ₹27,000 **(D)** ₹35,000

Q22. What is the interest on ₹10,000 for one year at 8% per annum compounded quarterly?

(A) ₹800.00 **(B)** ₹824.32 **(C)** ₹816.00 **(D)** ₹832.00

Part 7 — Mathematical Reasoning (Syllabus Chapter 10)

Q23. To prove $1\cdot2 + 2\cdot3 + \ldots + n(n+1) = n(n+1)(n+2)/3$ by induction, we assume it for $n = k$ and add the next term. What must $k(k+1)(k+2)/3 + (k+1)(k+2)$ be shown to equal?

(A) $(k+1)(k+2)/3$ **(B)** $k(k+1)(k+2)$

(C) $(k+2)(k+3)(k+4)/3$ **(D)** $(k+1)(k+2)(k+3)/3$

Q24. A corner square of an 8×8 chessboard is removed. For how many choices of a second square to remove could the remaining 62 squares possibly be covered by 31 dominoes, each covering two neighbouring squares?

(A) 31 **(B)** 63 **(C)** 32 **(D)** 62

Q25. Five points are placed inside or on a square of side 2. What is the largest distance d for which we can be sure that two of the points are at most d apart?

(A) 1 **(B)** 2 **(C)** $\sqrt{2}$ **(D)** $2\sqrt{2}$

Part 8 — Mixed Problems (each joins two chapters)

Q26. What is the least natural number N such that $n! > 2^n$ for every $n \geq N$?

(A) 4 **(B)** 3 **(C)** 5 **(D)** 2

Q27. In how many ways can 12 identical sweets be shared among 3 children so that each child gets at least 2?

(A) 55 **(B)** 91 **(C)** 45 **(D)** 28

Q28. Let $z = 1 + i$. What is $z + z^2 + z^3 + \ldots + z^8$?

(A) $15 + 15i$ **(B)** $15 - 15i$ **(C)** 16 **(D)** $-15 + 15i$

Q29. The line $\sqrt{3}x - y + 5 = 0$ is written as $x\cos\alpha + y\sin\alpha = p$ with $p > 0$. What are α and p?

(A) $\alpha = 150°$, $p = 5/2$ **(B)** $\alpha = 30°$, $p = 5/2$

(C) $\alpha = 150°$, $p = 5$ **(D)** $\alpha = 60°$, $p = 5/2$

Q30. The digits 1, 2, 3 and 4 are arranged in a random order to form a four-digit number. What is the probability that the number is divisible by 4?

(A) 1/2 **(B)** 1/4 **(C)** 1/3 **(D)** 1/6

Answer Sheet AMO Level 2 · Class 11 · Practice Paper 10

Name: **School:** **Roll No.:**

Shade or tick one circle for each question.

1.	A	B	C	D	11.	A	B	C	D	21.	A	B	C	D
2.	A	B	C	D	12.	A	B	C	D	22.	A	B	C	D
3.	A	B	C	D	13.	A	B	C	D	23.	A	B	C	D
4.	A	B	C	D	14.	A	B	C	D	24.	A	B	C	D
5.	A	B	C	D	15.	A	B	C	D	25.	A	B	C	D
6.	A	B	C	D	16.	A	B	C	D	26.	A	B	C	D
7.	A	B	C	D	17.	A	B	C	D	27.	A	B	C	D
8.	A	B	C	D	18.	A	B	C	D	28.	A	B	C	D
9.	A	B	C	D	19.	A	B	C	D	29.	A	B	C	D
10.	A	B	C	D	20.	A	B	C	D	30.	A	B	C	D

For the parent or teacher

Correct: _____ / 30

Answer Key and Solutions AMO Level 2 · Class 11 · Practice Paper 10

Quick answers

1 **A**	6 **B**	11 **C**	16 **D**	21 **A**	26 **A**
2 **D**	7 **C**	12 **C**	17 **C**	22 **B**	27 **D**
3 **C**	8 **B**	13 **A**	18 **D**	23 **D**	28 **B**
4 **A**	9 **B**	14 **C**	19 **D**	24 **C**	29 **A**
5 **A**	10 **D**	15 **B**	20 **B**	25 **C**	30 **B**

Solutions

Q	Part · level	Answer	Why?
1	Part 1 · Ch1 · Hard	**(A)** −135°	***Hint:*** *Expand: $(1 - i)^2 = -2i$.* $(1 - i)^3 = -2i(1 - i) = -2i + 2i^2 = -2 - 2i$. This is in the third quadrant with equal parts, so the principal argument is −135°. (225° is the same direction but outside the principal range.)
2	Part 1 · Ch1 · Hard	**(D)** 12	***Hint:*** *Expand $(1 - i)^2$ first, then compare real and imaginary parts.* $(1 - i)^2 = -2i$, so the left side is $2x + (x - 2y)i$. Real parts: $2x = 6$, so $x = 3$. Imaginary parts: $x - 2y = -5$, so $y = 4$. $xy = 12$.
3	Part 1 · Ch1 · Hard	**(C)** −64	***Hint:*** *The two terms are conjugates, so the sum is twice the real part of $(1 + i)^{11}$.* $(1 + i)^2 = 2i$, so $(1 + i)^8 = 16$ and $(1 + i)^{11} = 16(1 + i)^3 = 16(-2 + 2i)$. The real part is −32, so the sum is −64.
4	Part 2 · Ch4 · Hard	**(A)** 10	***Hint:*** *$S_n = 3(2^n - 1)$.* $3(2^n - 1) > 3000$ gives $2^n > 1001$. $2^9 = 512$ is too small and $2^{10} = 1024$ works, so $n = 10$.
5	Part 2 · Ch4 · Hard	**(A)** 1540	***Hint:*** *The kth term is $k(k + 1)/2$.* $\Sigma\, k(k + 1)/2 = \frac{1}{2}(\Sigma k^2 + \Sigma k) = \frac{1}{2}(2870 + 210) = 1540$. (Formula $n(n + 1)(n + 2)/6 = 20 \cdot 21 \cdot 22/6$.)
6	Part 2 · Ch4 · Hard	**(B)** 7	***Hint:*** *$2C(n, 2) = C(n, 1) + C(n, 3)$.* $n(n - 1) = n + n(n - 1)(n - 2)/6$. Dividing by n and simplifying: $n^2 - 9n + 14 = 0$, so $n = 7$ ($n = 2$ gives no 4th term). Check: 7, 21, 35.
7	Part 2 · Ch4 · Hard	**(C)** 1/3	***Hint:*** *Set the expression equal to k and require the quadratic in x to have real roots.* $k(x^2 + x + 1) = x^2 - x + 1$ gives $(k - 1)x^2 + (k + 1)x + (k - 1) = 0$. Real x needs $(k + 1)^2 - 4(k - 1)^2 \geq 0$, that is $(3k - 1)(3 - k) \geq 0$, so $1/3 \leq k \leq 3$. The least value is 1/3 (at $x = 1$).
8	Part 2 · Ch4 · Hard	**(B)** 12	***Hint:*** *Go through $x = 0, 1, 2, 3$ and list the allowed y.* $x = 0$: $4 \leq y \leq 5$ (2). $x = 1$: $2 \leq y \leq 4$ (3). $x = 2$: $0 \leq y \leq 3$ (4). $x = 3$: $0 \leq y \leq 2$ (3). Total 12.
9	Part 2 · Ch4 · Hard	**(B)** 6	***Hint:*** *Use $(a + b + c)^2$ and $b^2 = ac$.* $ab + bc + ca = (26^2 - 364)/2 = 156$. Since $ac = b^2$, this is $b(a + c) + b^2 = b(26 - b) + b^2 = 26b$. So $b = 6$ (the numbers are 2, 6, 18).
10	Part 3 · Ch5 · Hard	**(D)** $2x + y = 1$	***Hint:*** *A point (x, y) reflects in $x = 1$ to $(2 - x, y)$.* Replace x by $2 - x$: $2(2 - x) - y = 3$, so $2x + y = 1$. Check: $(0, -3)$ on the line goes to $(2, -3)$, and $4 - 3 = 1$.

Q	Part · level	Answer	Why?
11	Part 3 · Ch5 · Hard	**(C)** 14/3	***Hint:*** *Area = ½|x₁(y₂ − y₃) + x₂(y₃ − y₁) + x₃(y₁ − y₂)|.* Area = ½|1(3 − 7) + k(7 − 1) + 5(1 − 3)| = ½|6k − 14| = 4. So 6k − 14 = ±8, giving k = 11/3 or 1. Sum = 14/3. (k = 7/3 makes the points collinear.)
12	Part 3 · Ch5 · Hard	**(C)** 12	***Hint:*** *With intercepts a and b, 2/a + 3/b = 1. Use AM–GM on 2/a and 3/b.* 1 = 2/a + 3/b ≥ 2√(6/(ab)), so ab ≥ 24 and the area ab/2 ≥ 12. Equality when 2/a = 3/b = 1/2: a = 4, b = 6, when (2, 3) is the midpoint of AB.
13	Part 4 · Ch7 · Hard	**(A)** 7	***Hint:*** *Write 3 cos x + 4 sin x as 5 cos(x − α) for a suitable angle α.* 3 cos x + 4 sin x = 5(3/5 cos x + 4/5 sin x) = 5 cos(x − α) with cos α = 3/5, so it is at most 5. The greatest value is 5 + 2 = 7. (9 adds 3 + 4 + 2.)
14	Part 4 · Ch7 · Hard	**(C)** 7/8	***Hint:*** *$\sin^4\theta + \cos^4\theta = (\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta$.* $\sin^4\theta + \cos^4\theta = 1 - \frac{1}{2}\sin^2 2\theta$. With θ = 15°: $1 - \frac{1}{2} \times (1/2)^2 = 1 - 1/8 = 7/8$.
15	Part 4 · Ch7 · Hard	**(B)** √2 − 1	***Hint:*** *Use tan(θ/2) = (1 − cos θ)/sin θ with θ = 45°.* tan 22.5° = (1 − cos 45°)/sin 45° = (1 − 1/√2)/(1/√2) = √2 − 1. (2 − √3 is tan 15°.)
16	Part 5 · Ch8 · Hard	**(D)** 240	***Hint:*** *First choose 4 of the 6 couples, then one person from each.* Choose 4 couples: C(6, 4) = 15. From each chosen couple pick one of 2 people: $2^4 = 16$. Total 240.
17	Part 5 · Ch8 · Hard	**(C)** 30	***Hint:*** *Use the sum of the observations and the sum of their squares.* Sum = 5 × 6 = 30, so the other two add to 30 − 18 = 12. Variance = (sum of squares)/5 − 6^2 = 4, so the sum of squares is 5 × 40 = 200. The known squares give 16 + 36 + 64 = 116, so $a^2 + b^2 = 84$. Then $2ab = (a + b)^2 - (a^2 + b^2) = 144 - 84 = 60$, so ab = 30. (36 assumes both are 6, which gives variance 1.6.)
18	Part 5 · Ch8 · Hard	**(D)** 16/27	***Hint:*** *Add P(3) and P(4).* $P(3) = 4 \times (2/3)^3 \times (1/3) = 32/81$. $P(4) = (2/3)^4 = 16/81$. Total 48/81 = 16/27.
19	Part 5 · Ch8 · Hard	**(D)** 4	***Hint:*** *Find the variance of x first. Adding 3 changes nothing; doubling doubles the SD.* Mean = 6 and mean of squares = 40, so variance = 40 − 36 = 4 and SD = 2. The SD of 2x + 3 is 2 × 2 = 4. (16 is the variance.)
20	Part 6 · Ch9 · Hard	**(B)** More than 75 days	***Hint:*** *Solve 30,000 + 100d < 500d.* 30,000 + 100d < 500d gives 400d > 30,000, so d > 75. Buying is cheaper for more than 75 days.
21	Part 6 · Ch9 · Hard	**(A)** ₹32,000	***Hint:*** *Take out the salary, then share the rest.* Salary = ₹5,000. Remaining ₹45,000 is shared 3 : 2, so Kiran gets ₹27,000 more. Total ₹32,000.
22	Part 6 · Ch9 · Hard	**(B)** ₹824.32	***Hint:*** *The rate per quarter is 2%, applied 4 times.* Amount = $10{,}000 \times 1.02^4 = 10{,}000 \times 1.08243216$ = ₹10,824.32. Interest = ₹824.32. (Half-yearly would give ₹816.)
23	Part 7 · Ch10 · Hard	**(D)** (k + 1)(k + 2)(k + 3)/3	***Hint:*** *The case n = k + 1 replaces n by k + 1 in the formula.* The next term is (k + 1)(k + 2). k(k + 1)(k + 2)/3 + (k + 1)(k + 2) = (k + 1)(k + 2)(k + 3)/3, which is the formula for n = k + 1.

Q	Part · level	Answer	Why?
24	Part 7 · Ch10 · Hard	**(C)** 32	***Hint:*** *Each domino covers one black and one white square.* The corner is, say, black, leaving 31 black and 32 white squares. Dominoes need equal numbers, so the second square must be white: 32 choices. (In fact every such choice can be tiled.)
25	Part 7 · Ch10 · Hard	**(C)** √2	***Hint:*** *Cut the square into four squares of side 1.* Four unit squares and five points: two points share a unit square (pigeonhole), and its diagonal is √2. Points at the four corners and the centre have smallest gap exactly √2, so no smaller d is guaranteed.
26	Part 8 · Mixed · Ch4 + Ch10 · Hard	**(A)** 4	***Hint:*** *Check n = 3 and n = 4, then the step: (k + 1)! = (k + 1)k! .* $3! = 6 < 8 = 2^3$, but $4! = 24 > 16 = 2^4$. If $k! > 2^k$ with $k \geq 4$, then $(k + 1)! = (k + 1)k! > 2 \times 2^k = 2^{k+1}$. So N = 4.
27	Part 8 · Mixed · Ch4 + Ch8 · Hard	**(D)** 28	***Hint:*** *Give each child 2 first, then share the remaining 6 freely.* After giving 2 each, 6 sweets remain for 3 children with no restriction: C(6 + 2, 2) = C(8, 2) = 28.
28	Part 8 · Mixed · Ch1 + Ch4 · Hard	**(B)** 15 − 15i	***Hint:*** *It is a GP with ratio z, and $z^8 = 16$.* Sum = $z(z^8 - 1)/(z - 1) = (1 + i)(16 - 1)/i = 15(1 + i)/i = 15(1 + i)(-i) = 15(1 - i) = 15 - 15i$.
29	Part 8 · Mixed · Ch5 + Ch7 · Hard	**(A)** α = 150°, p = 5/2	***Hint:*** *Move the constant to the right, make it positive, and divide by √(3 + 1) = 2.* $-\sqrt{3}x + y = 5$. Dividing by 2: $(-\sqrt{3}/2)x + (1/2)y = 5/2$. So cos α = $-\sqrt{3}/2$ and sin α = 1/2, giving α = 150°, and p = 5/2.
30	Part 8 · Mixed · Ch8 + Ch10 · Hard	**(B)** 1/4	***Hint:*** *A number is divisible by 4 when its last two digits form a multiple of 4.* The last two digits must be 12, 24 or 32 (the other two-digit endings from these digits are not multiples of 4). Each leaves 2 orders for the first two digits: 3 × 2 = 6 numbers out of 4! = 24. P = 6/24 = 1/4.